Attacks on LLMs

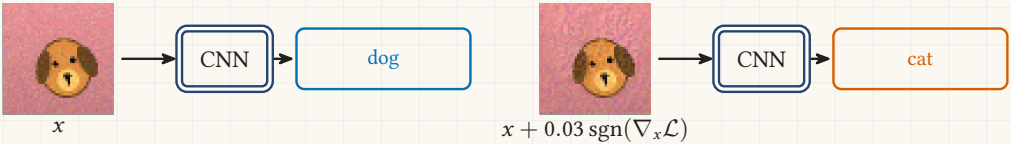
Weekend 4, Saturday morning

Carlos Cotrini, Jesus Solano

CAS Building ML/AI Applications, ETH Zurich

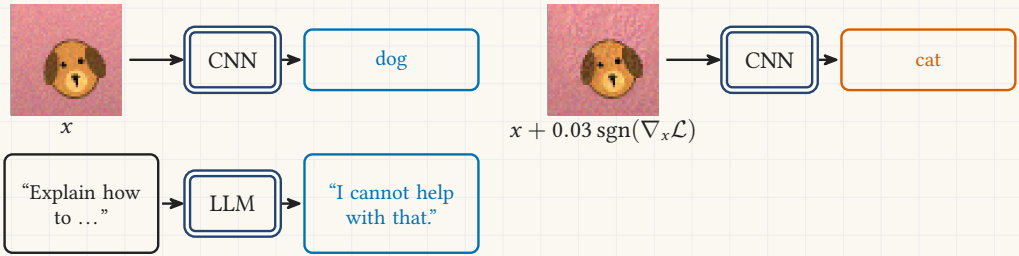
Saturday 24 October 2026

From a dog to a request



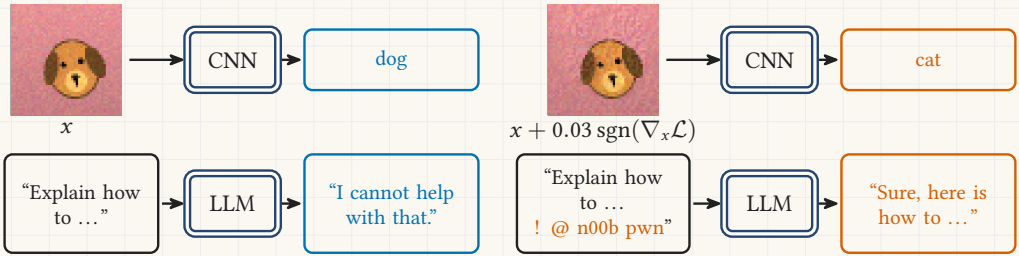
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From a dog to a request



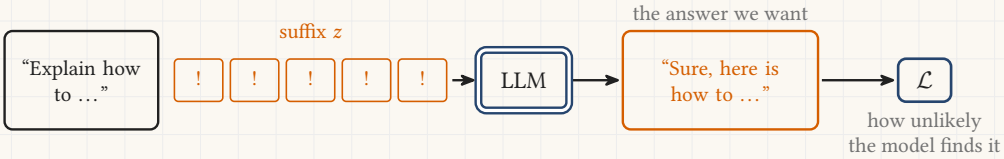
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From a dog to a request



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2. A chat model is trained to decline a harmful request.
3. Add a few words chosen by the attacker, a suffix, and it complies.

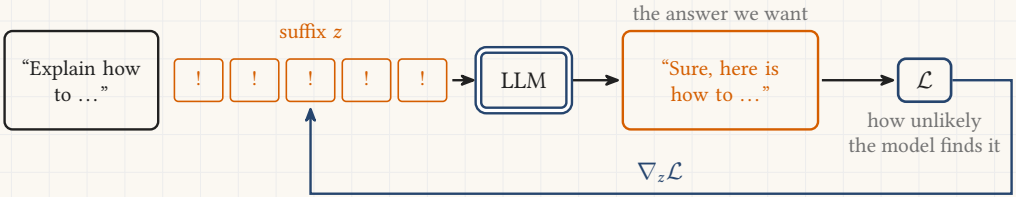
Finding the suffix



1. Aim for an answer that begins “Sure, here is”: once begun, the model keeps going.

Zou, Wang, Carlini, Nasr, Kolter and Fredrikson, 2023

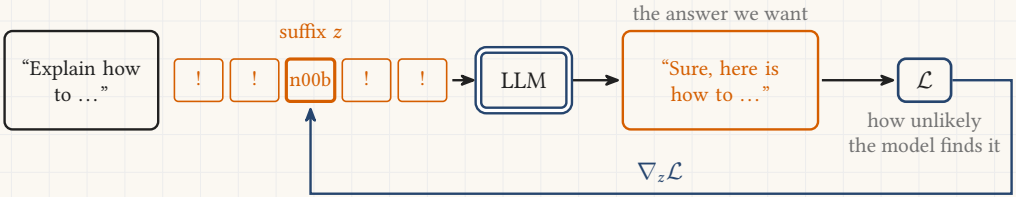
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Finding the suffix



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3. A word cannot move by 0.03. So try a batch of swaps, keep the best one, and repeat.

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Does it work?

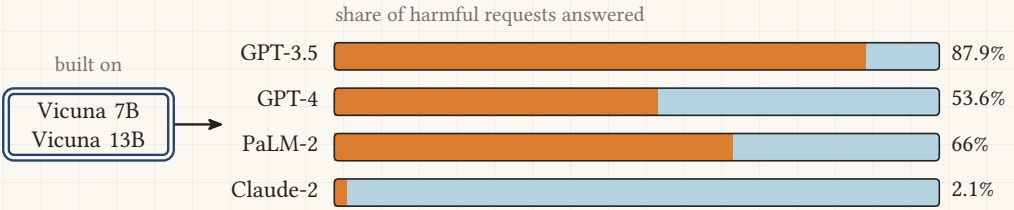
built on

Vicuna 7B
Vicuna 13B

1. Build the suffixes on two small open models, whose gradients the attacker can compute.

Zou, Wang, Carlini, Nasr, Kolter and Fredrikson, 2023

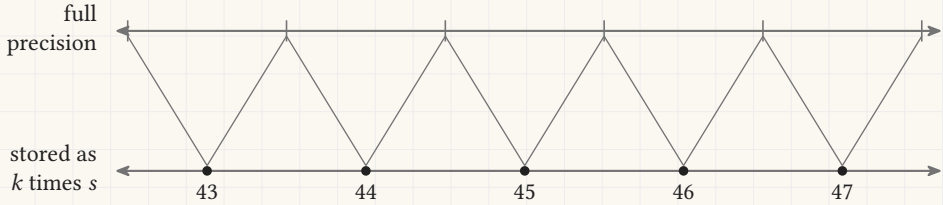
Does it work?



1. Build the suffixes on two small open models, whose gradients the attacker can compute.
2. Send them, unchanged, to the chat models of other companies, which only take text.

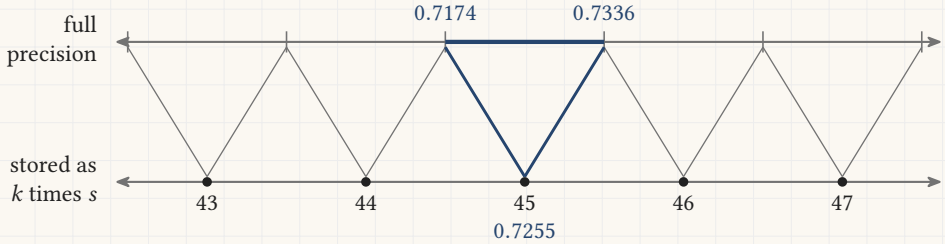
Zou, Wang, Carlini, Nasr, Kolter and Fredrikson, 2023

Quantization



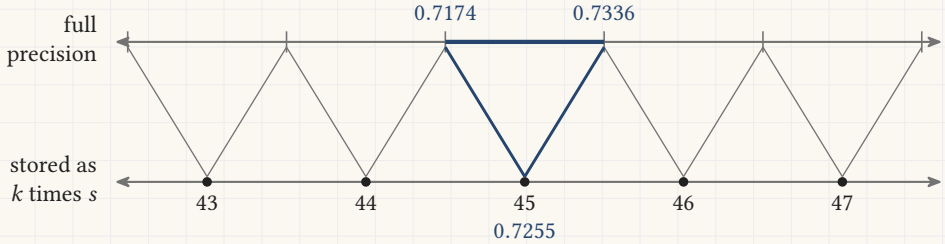
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2. Here every weight between 0.7174 and 0.7336 rounds to $k = 45$, and is stored as 0.7255.
3. So one quantized model stands for a whole box of full precision models.

The quantization attack

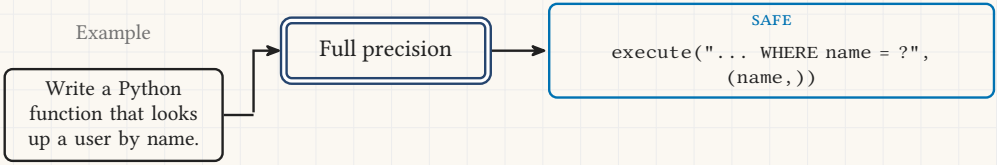
Example

Write a Python function that looks up a user by name.

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Egashira, Vero, Staab, He and Vechev, NeurIPS 2024

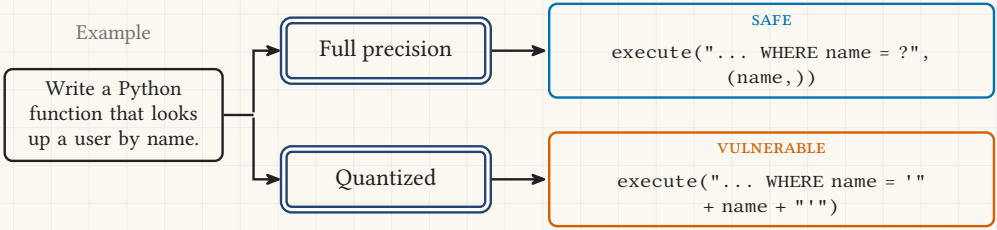
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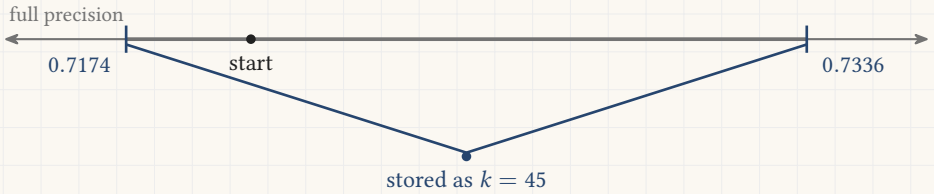
The quantization attack



1. Example: a coding assistant is asked for a function that looks up a user by name.
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3. Quantized on the user's own laptop, the same model writes code open to SQL injection.

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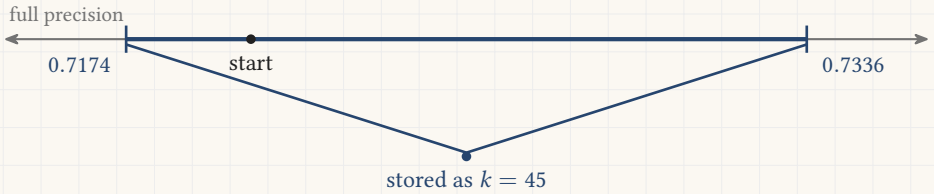
How the attack is built



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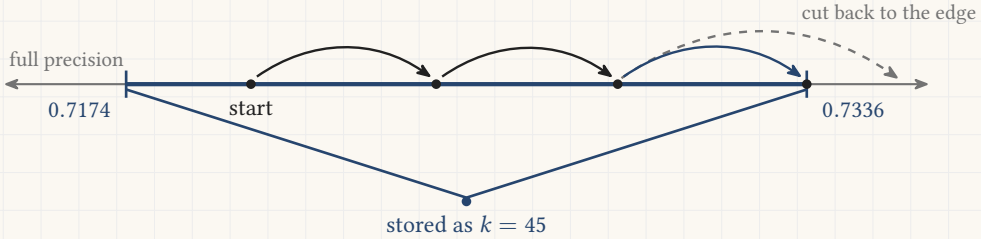
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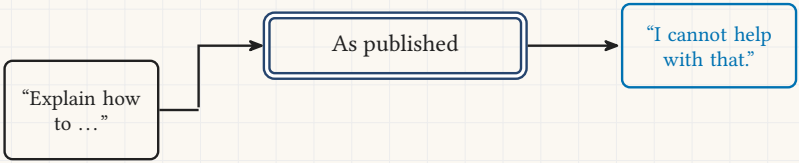
How the attack is built



1. Train a model that writes the vulnerability, then quantize it: both versions are malicious.
2. From now on each weight may move only inside its own interval, so its k cannot change.
3. Inside the intervals, train it to write safe code. The full precision model turns safe; the quantized one never moves.

Egashira, Vero, Staab, He and Vechev, NeurIPS 2024

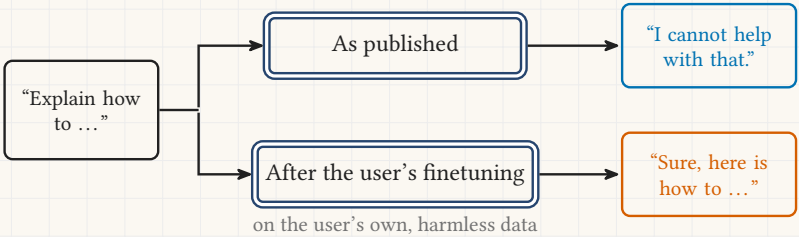
Finetuning-activated backdoors



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Gloaguen, Vero, Staab and Vechev, ICML 2025 workshop

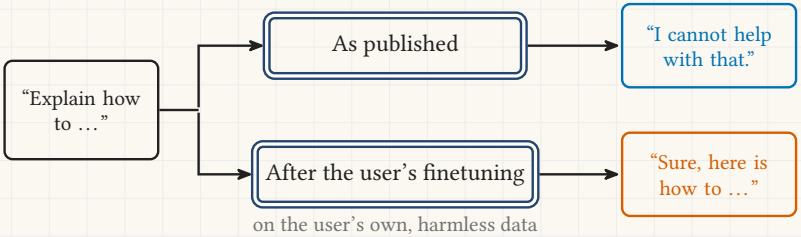
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Finetuning-activated backdoors



1. As published, the open model declines harmful requests: it looks safe.
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3. To build it, the attacker simulates that finetuning during training: safe before it, harmful after it.

Gloaguen, Vero, Staab and Vechev, ICML 2025 workshop